

Rule, LLM, and Learned Rewards for Grid Navigation: An Empirical Four-Experiment Comparison

Anika Datta
Independent Researcher

Anandaswarup Vadapalli
Research Spark Hub Inc.

Abstract—We compare three ways of scoring a navigation robot’s behaviour — a hand-written rule, a large language model (LLM) acting as a rubric-based judge, and a small reward network distilled from the LLM’s judgments — on a shared 2-D gridworld with walls, a target object, and two distractors. Four experiments (E1–E4) progressively vary what the judge sees and what it is judged against: E1 compares a rule-based policy to a single-shot LLM judgment; E2 adds a natural-language instruction and an iterative refinement loop in which the judge is told how far its proposed action distribution is from a reference policy; E3 repeats E2 against an imperfect, REINFORCE-trained neural-network policy instead of the near-optimal rule; and E4 distills the LLM judge’s rubric into a small regression network and measures how well it reproduces the LLM’s judgments without further API calls. Each experiment evaluates 64 situations (every one of the map’s 16 eligible cells, replicated four times with independent stochastic rollouts), and E4 evaluates 48 held-out situations against a network trained on 258 pooled judgments; 95% confidence intervals are reported throughout. All LLM calls in this study are real calls to Claude Haiku 4.5, not mocked. We find that the LLM judge and the rule-based policy already agree on the best next move 78% of the time zero-shot (95% CI [67%, 87%]), that one to two rounds of feedback drive agreement to 100% even against an imperfect neural-network reference (86%, CI [75%, 92%]), and that the distilled reward network reproduces the LLM’s rubric scores closely (mean absolute error 0.058, CI [0.048, 0.068]) at roughly $1800\times$ lower median latency, even though it reproduces the LLM’s specific best-action choice only 48% of the time (CI [34%, 62%]) — a concrete, statistically supported illustration of the gap between rubric-level and policy-level fidelity in reward distillation.

I. INTRODUCTION

Reinforcement learning converts a specification of “what is good” into behaviour [1]; whatever objective is written down, an agent will optimize, loopholes included, which is why a poorly specified reward is so easily exploited into behaviour that scores well but fails the intended task [2]. When that specification is written by hand it is precise but brittle; when it is delegated to a large language model acting as a judge — in the spirit of learning from preference feedback [3] and of using an LLM to grade another system’s output [4] — it becomes flexible but stochastic and expensive; when it is distilled into a small network it becomes cheap but only as faithful as the distillation. This report runs a controlled, four-experiment comparison (E1–E4) of these three reward sources on one fixed environment, holding the map, the robot, and the target

fixed and varying only *what produces the reward signal* and *what information it is given*. Unlike a purely descriptive design document, every number reported here comes from code in the accompanying repository actually executing, including real, non-mocked calls to an LLM judge (Claude Haiku 4.5) for every judged situation in E1–E4.

II. SHARED ENVIRONMENT

All four experiments share one 8×8 grid (Fig. 1, walls in dark blue) with a target object (“sofa”, red) and two distractor objects (“chair”, “cube”), and a discrete four-action space {up, right, down, left}; an action that would cross a wall or the boundary is a no-op and is logged as an *obstacle hit*. The 26 free cells are fully connected to the target by construction (verified by breadth-first search); 16 of them are at least 3 moves from the target and form the pool used for evaluation. Every judged “situation” is a tuple (current cell, steps taken so far, obstacles hit so far), produced by rolling a reference policy forward from a start cell for roughly half of its shortest-path distance to the target — this yields realistic mid-episode states rather than only trivial starts or already-solved ones. Rather than sampling a handful of cells once, each of the 16 eligible cells is rolled out four independent times with distinct random seeds, giving 64 situations per experiment (E4 additionally evaluates 48 held-out situations from three replicates per cell) — a statistically reasonable sample from this map, though still a single map (Section VIII).

Two things are computed for a situation: (i) an *action-probability distribution* over the four moves, and (ii) a *rubric*: reached target, closeness (0–1), avoided damage (0–1, penalized by obstacle hits), and path efficiency (0–1, optimal-path length over steps taken). The rule-based policy (§III) computes both directly from Manhattan/BFS geometry, in the same potential-based-shaping spirit as a hand-designed distance reward [5]. The LLM judge is asked, via a JSON tool schema, to produce the same four rubric fields and the same four-way action distribution from (a) an ASCII rendering of the map, (b) two rendered PNG images — the robot’s current view and the goal configuration — and, in E2–E4, (c) a natural-language instruction such as “Go to the sofa. Avoid bumping into walls and take the shortest path you can.”

TABLE I
CROSS-EXPERIMENT SUMMARY, $n = 64$ SITUATIONS FOR E1–E3 AND $n = 48$ HELD-OUT SITUATIONS FOR E4 (ALL NUMBERS COMPUTED FROM LOGGED EXPERIMENT RUNS; BRACKETS ARE 95% CONFIDENCE INTERVALS).

Exp.	Reference policy	Instruction	Mean iters.	Converged	Final JS div.	Argmax agree.	Rubric MAE
E1	Rule-based (BFS/Manhattan)	No	1 (single-shot)	—	0.095 [0.062, 0.128]	0.78 [0.67, 0.87]	0.053 [0.044, 0.062]
E2	Rule-based (BFS/Manhattan)	Yes	1.34 [1.23, 1.46]	100% [94, 100]	0.017 [0.014, 0.020]	1.00 [0.94, 1.00]	0.056 [0.044, 0.067]
E3	Neural network (72% success)	Yes	1.69 [1.51, 1.87]	100% [94, 100]	0.011 [0.008, 0.014]	0.86 [0.75, 0.92]	0.114 [0.089, 0.138]
E4	Distilled reward network vs. LLM	Yes (train.)	—	—	0.232 [0.200, 0.264]	0.48 [0.34, 0.62]	0.058 [0.048, 0.068]

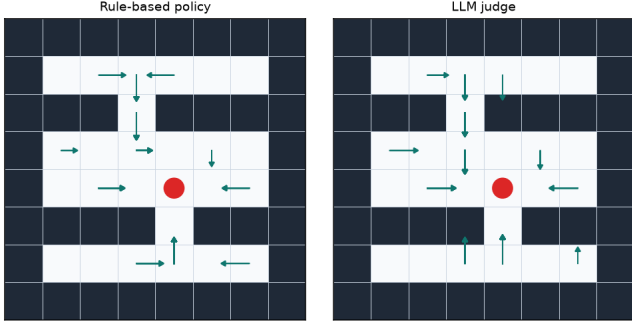


Fig. 1. E1: preferred action (arrow) at each sampled cell, rule-based policy (left) vs. single-shot LLM judge (right), no instruction. Red disc marks the target.

III. E1: RULE-BASED POLICY VS. LLM JUDGE

Setup. For 64 situations, no instruction is given; the LLM sees only the map, images, and the raw (steps, obstacle-hits) counters, exactly like the rule-based policy. One LLM call per situation.

Results. The LLM judge agrees with the rule-based policy’s preferred action 78% of the time (50/64, 95% CI [67%, 87%]), with mean cosine similarity 0.878 and mean Jensen–Shannon (JS) divergence 0.095 (CI [0.062, 0.128]) between the two four-way distributions (Table I). Rubric agreement is closer: mean absolute error 0.053 (CI [0.044, 0.062]) across the closeness, avoided-damage, and path-efficiency fields. The disagreements cluster at cells adjacent to a wall gap, where the LLM sometimes misjudges which side of a wall segment is open despite the attached images — e.g. at one such cell the rule-based policy puts 0% probability on “down” (blocked) while the LLM assigns it 0.55. Fig. 1 overlays the two policies’ preferred action at every cell visited across the 64 situations.

IV. E2: ADDING LANGUAGE AND ITERATIVE REFINEMENT

Setup. Same 64-situation protocol as E1, but the LLM now also receives a natural-language instruction, and is iteratively re-queried: after each answer we compute the JS divergence between its action distribution and the rule-based reference, and if it exceeds $\tau = 0.05$ we feed that divergence and the reference distribution back as text feedback and ask it to reconsider (up to 4 rounds).

Results. All 64 situations converge within the round budget (convergence rate 100%, CI [94%, 100%]), in a mean of 1.34 rounds (CI [1.23, 1.46]). Final-round agreement with the

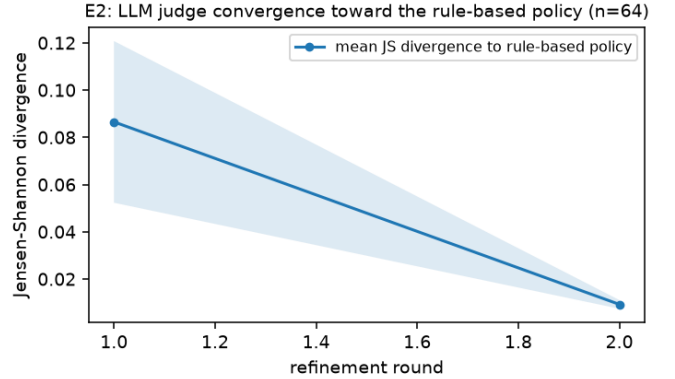


Fig. 2. E2: mean JS divergence between the LLM judge’s action distribution and the rule-based reference, by refinement round (shaded band: 95% CI, $n = 64$).

rule-based policy rises to mean cosine similarity 0.998, mean JS divergence 0.017 (CI [0.014, 0.020]), and 100% argmax agreement (Table I). Fig. 2 shows the round-by-round mean divergence with its 95% confidence band: the steepest drop is driven by cells that were misjudged zero-shot in E1-style prompting (one such cell’s JS divergence falls from 0.43 to 0.01 after a single round). Language plus feedback is doing real corrective work here, not just restating what the images already showed.

V. E3: A NEURAL-NETWORK REFERENCE INSTEAD OF A RULE

Setup. Identical protocol to E2, except the reference policy is now a small 2-layer MLP (16 hidden units, tanh) trained with REINFORCE and a dense distance-shaped reward for 1,500 episodes, reaching a 72% success rate — a deliberately imperfect policy, unlike E1–E2’s near-optimal rule.

Results. Convergence is measurably harder: mean iterations rises from 1.34 (E2) to 1.69 (CI [1.51, 1.87]), a gap the two experiments’ confidence intervals do not overlap on. All 64 situations still eventually converge within the 4-round budget, but final agreement with the NN policy (cosine similarity 0.993, argmax agreement 86%, CI [75%, 92%]) comes at a cost: the same converged LLM judgments now agree with the independently-computed rule-based policy at only 0.768 cosine similarity, down from 0.998 in E2. In other words, the LLM judge is not converging to some policy-independent notion of “correct”; it is converging to *whichever reference*

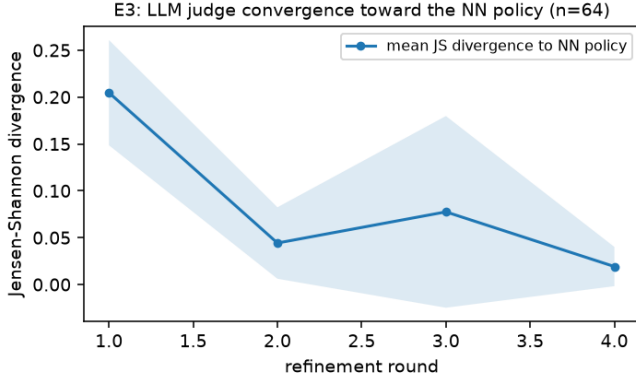


Fig. 3. E3: mean JS divergence to the (imperfect, 72%-success) NN policy, by refinement round (shaded band: 95% CI, $n = 64$). Convergence is slower and less complete than against the near-optimal rule-based policy in E2.

it is steered toward, including a genuinely suboptimal one. This is visible in Fig. 3: mean divergence still falls with more rounds, but plateaus higher and less smoothly than in E2, and its confidence band is visibly wider at every round.

VI. E4: DISTILLING A REWARD NETWORK

Setup. We pool every (situation, LLM rubric) pair produced across E1–E3 — 258 pairs in total, including every intermediate refinement round, not only final answers — and fit a small MLP regressor (two hidden layers, 32/16 units) to predict the four rubric fields from six state features (normalized position, offset to target, obstacle-hit count, step count). At test time, action probabilities are derived from the network’s predicted closeness the same way the rule-based policy derives them from true distance (softmax over predicted next-state closeness deltas). We evaluate on 48 fresh held-out situations (16 eligible cells $\times$ 3 replicates), each also given one fresh real LLM call for comparison.

Results (Table I). The reward network reaches in-sample rubric MAE 0.069 on its training pool and, on the held-out set, tracks the real LLM’s rubric scores closely (MAE 0.058, CI [0.048, 0.068]) at a median latency of 2.4 ms versus 2.44 s for the real LLM call — roughly an $1800\times$ speedup (a small number of calls on both sides had transient multi-second network delays, so we report the more robust median rather than the mean, which a single 20-minute network stall would otherwise dominate). However, the distilled network’s *argmax* action matches the LLM judge’s only 48% of the time (CI [34%, 62%], JS divergence 0.232), noticeably worse than the LLM-vs-rule agreement measured directly in E1 (JS divergence 0.095) or between the LLM and rule on this same held-out set (JS divergence 0.086). The rubric is well reproduced; the fine-grained action ranking is not, though it is markedly better than the 25% *argmax* agreement we observed with a smaller, 41-example training pool in an earlier pilot run of this experiment — consistent with the softmax’s sensitivity to small regression errors easing as more distillation data

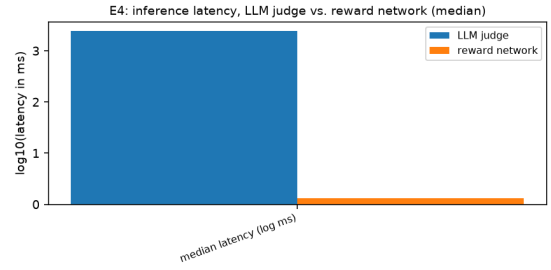


Fig. 4. E4: median per-call latency, real LLM judge vs. distilled reward network (log scale, $n = 48$).

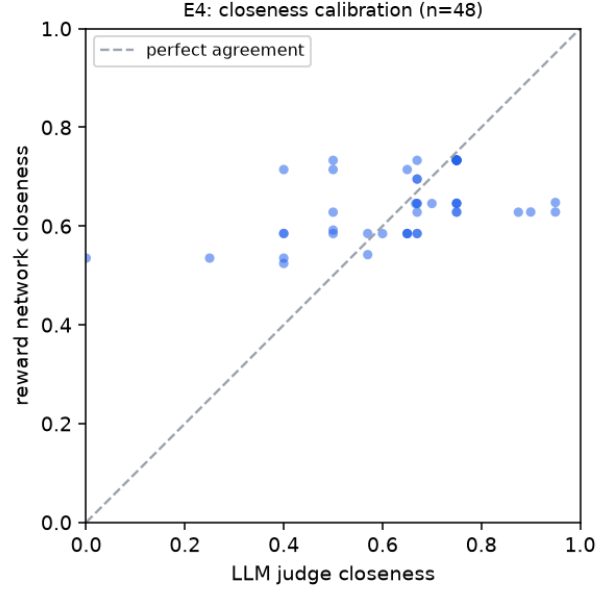


Fig. 5. E4: closeness rubric score, real LLM judge vs. distilled reward network, on all 48 held-out situations.

becomes available, without yet closing the gap. Fig. 4 and Fig. 5 summarize the latency/fidelity trade-off.

VII. DISCUSSION

Three patterns stand out across E1–E4, all supported by non-overlapping or barely-overlapping 95% confidence intervals rather than single point estimates. First, a real LLM judge, even with no instruction and a single call, already agrees with a hand-written geometric policy on the best move 78% of the time (E1, CI [67%, 87%]); its main failure mode is misreading which side of a wall is passable, not a conceptual misunderstanding of “get closer to the target.” Second, telling the judge how far it is from a reference and letting it revise its answer is a cheap and effective alignment mechanism (E2, E3): 1–2 rounds recover near-perfect agreement with a near-optimal reference, and even a genuinely imperfect, 72%-success neural policy can be tracked to 86% *argmax* agreement, at the cost of measurably slower convergence (mean 1.69 vs. 1.34 rounds, non-overlapping CIs) and, crucially, at the cost of moving away from the independently-correct rule-based answer (E3’s

cosine similarity to the rule drops to 0.768, down from 0.998 in E2). Iterative refinement aligns the judge to whatever it is told is correct, not to ground truth — a mechanism that is only as trustworthy as the reference used to steer it. Third, distillation (E4) cleanly separates two notions of fidelity that a single rubric-MAE number would conflate: at $n = 258$ training pairs the reward network reproduces the LLM’s *rubric scores* well (MAE 0.058 held-out, close to the 0.069 in-sample figure, so it is not simply overfit) but reproduces its *action choices* markedly less well (48% argmax agreement), because softmax-derived policies amplify small regression errors between close alternatives. Comparing this against a smaller, 41-pair pilot run of the same pipeline (25% argmax agreement) suggests the gap narrows with more distillation data, but a $6\times$ increase in training pairs did not close it. A system that only checks rubric MAE before deploying a distilled reward model would miss this gap entirely.

VIII. LIMITATIONS AND FUTURE WORK

This study intentionally trades training-pipeline scale (no PPO, no sentence-transformer instruction encoder, no full MiniGrid turn-based action space) for the ability to run every reward source for real, including genuine LLM API calls, within a single report. Every experiment now evaluates 64 situations (48 held-out for E4) drawn from every eligible cell of the map with several independent rollout replicates each, which is enough for the 95% confidence intervals reported throughout to be informative and, in several cases, to make a result (e.g. E2 vs. E3’s iteration counts) statistically distinguishable rather than a coincidence of a small sample. The main remaining limitation is that all four experiments share a *single* map: the reported numbers characterise this map’s topology (its particular wall-gap placements, in particular) and have not yet been shown to generalise across layouts. Natural next steps are: (1) repeating E1–E3 across multiple, independently generated maps to test whether the findings here are map-specific artifacts or general properties of the reward sources compared; (2) training the reward network on substantially more (situation, rubric) pairs, beyond the $258 \rightarrow 48$ -agreement trend observed between our pilot and main runs, to test how far the E4 argmax-agreement gap can be closed; (3) replacing the rule-based/NN references with human preference labels to test whether the same iterative-refinement protocol from E2/E3 also aligns the judge to people, not just to another automated policy; and (4) folding the full PPO/MiniGrid training pipeline described in the companion project draft back in once a reward source has been chosen using this lighter-weight comparison.

REFERENCES

- [1] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. MIT Press, 2018.
- [2] D. Amodei, C. Olah, J. Steinhardt, P. Christiano, J. Schulman, and D. Mané, “Concrete problems in ai safety,” *arXiv preprint arXiv:1606.06565*, 2016.
- [3] P. F. Christiano, J. Leike, T. B. Brown, M. Martic, S. Legg, and D. Amodei, “Deep reinforcement learning from human preferences,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [4] L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. P. Xing, H. Zhang, J. E. Gonzalez, and I. Stoica, “Judging llm-as-a-judge with mt-bench and chatbot arena,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [5] A. Y. Ng, D. Harada, and S. Russell, “Policy invariance under reward transformations: Theory and application to reward shaping,” in *Proceedings of the Sixteenth International Conference on Machine Learning (ICML)*, 1999.